

FARI

Example Consolidated Report

Pilot Validation Portfolio

Release source: FARI-CONSOLIDATED-ASSESSMENT-REPORT.md

FARI v1.2.0 | 2026-06-26

DOES NOT MEET

Qualitative executive conclusion from the example report

This is the versioned release edition of the FARI documentation bundle. Canonical working sources for this release are stored under the same version folder.

Document Control

- **Assessment ID:** ASM-FARI-2026-VAL-001
- **Report version:** 0.3
- **Report maturity:** Provisional
- **Report date:** 2026-06-08
- **Assessment owner:** FARI framework validation owner
- **Report author:** FARI Report Author
- **Source material:** cases/01_Lib.md, cases/02_USB.md, cases/03_USB.md, and cases/04_Spacecanbus.md
- **Declared gating claims:** CLM-LIB-001, CLM-USB-001, CLM-USB-002, and CLM-SCAN-001

1. Overall FARI Result Card

Field	Result
Overall Conclusion	Does Not Meet
Confidence	High for the demonstrated library failure, medium for the SpaceCAN lab failure, and low for total infrastructure coverage
Required Action	Remediate demonstrated failures, Extend Investigation of both USB scenarios, then Retest
Priority	Immediate before affected components or displayed bus state are approved for mission-critical use
Scope Boundary	Only the four supplied pilot cases and evidenced environments; this is not a complete infrastructure or spacecraft assessment

Conclusion rationale: All four claims are gating for this pilot portfolio. CLM-LIB-001 and CLM-SCAN-001 are Does Not Meet; therefore the overall conclusion is Does Not Meet. Both USB claims remain Inconclusive and prevent a complete assurance statement, but they do not override demonstrated claim failures.

2. Executive Decision Summary

The tested OSDLP telecommand library revision and the evidenced SpaceCAN monitoring behavior must not be approved for mission-critical use until their demonstrated claim failures are remediated and retested. The two USB investigations require controlled, instrumented execution before a pass/fail statement can be made.

This report consolidates the portfolio so an executive can see the totaled conclusion, the decisions it drives, and the unresolved evidence boundaries without reviewing four technical reports first.

3. Investigation and Claim Results

Investigation	Claim	Gating	Conclusion	Confidence	Required Action	Report
INV-LIB-001	CLM-LIB-001: malformed telecommand inputs do not cause unsafe memory access or termination	Yes	Does Not Meet	High in tested host scope	Remediate, then Retest	01-library-fuzzing-report.md
INV-USB-001	CLM-USB-001: unsafe beacon intervals are rejected without mission-loop degradation	Yes	Inconclusive	Low	Extend Investigation	02-usb-interface-report.md
INV-USB-002	CLM-USB-002: undersized broadcast telecommands are rejected without unsafe memory access or command loss	Yes	Inconclusive	Low	Extend Investigation, then Retest	03-usb-apid-underflow-report.md
INV-SCAN-001	CLM-SCAN-001: forged or replayed SpaceCAN state is rejected or visibly distinguished before trusted use	Yes	Does Not Meet	Medium in lab monitor scope	Remediate, then Retest	04-spacecanbus-spoofing-replay-report.md

4. Totalized Results

Conclusion	Gating Claims	Open Actions
Meets	0	0
Does Not Meet	2	4
Inconclusive	2	2
Not Assessed	0	0

The counts support navigation and workload planning. They are not averaged into a vulnerability, mission-risk, or security score.

5. Required Decisions and Actions

Action	Related Result	Priority	Acceptance Criteria
Correct minimum-length handling in OSDLP and preserve regression inputs	INV-LIB-001	Immediate	Supplied crash inputs and expanded malformed-input suite complete without sanitizer findings or abnormal termination
Validate the patched OSDLP library in representative integration	INV-LIB-001	Immediate before integration approval	Integration test resolves reachability, recovery, and mission-consequence assumptions
Execute the controlled USB beacon-rate reproduction with corrected command encoding	INV-USB-001	Planned	Exact command bytes, acknowledgement, observed behavior, resource measurements, and recovery are captured
Instrument and retest the APID 0x06 undersized packet path	INV-USB-002	Immediate before resilience is claimed	Handler entry, rejection or unsafe effect, post-test command behavior, and recovery are captured
Add SpaceCAN origin/freshness controls and anomaly handling	INV-SCAN-001	Immediate before trusted operational use	Forged and replayed frames are rejected or visibly marked untrusted
Retest every representative SpaceCAN consumer	INV-SCAN-001	Immediate before operational approval	Monitor, control, autonomy, alarm, and recovery behavior meet defined acceptance criteria

6. Coverage and Limitations

Area	Coverage	Consolidated Result
OSDLP telecommand receive/unpack paths	Tested	Does Not Meet
USB beacon-rate scenario	Reviewed; execution not evidenced	Inconclusive
USB APID 0x06 undersized packet	Sender-side execution evidenced; target effect not evidenced	Inconclusive
SpaceCAN injection and lab monitor display	Tested	Does Not Meet
SpaceCAN replay transmission	Tested; receiver effect partial	Candidate replay-control failure
Integrated flight software and physical subsystems	Not assessed	No conclusion
RF delivery and authentication	Not assessed	No conclusion
Hardware and on-orbit behavior	Not assessed / out of scope	No conclusion
Remaining infrastructure	Not assessed	No conclusion

The overall Does Not Meet conclusion applies only to the declared pilot portfolio. It must not be presented as a conclusion about an entire spacecraft, mission, or client infrastructure.

7. Consolidated Assurance Statement

The FARI pilot validation portfolio **Does Not Meet** its declared gating claims because claim failures are directly demonstrated in the tested OSDLP library and SpaceCAN lab-monitor scopes. Both USB claims are Inconclusive and require extended investigation. No conclusion is made about the remainder of any client infrastructure, physical spacecraft behavior, RF reachability, or on-orbit consequence.

8. Closing FARI Result Card

Field	Result
Overall Conclusion	Does Not Meet
Confidence	Mixed: high for OSDLP host failure, medium for SpaceCAN lab failure, low for portfolio completeness
Required Action	Remediate, Extend Investigation, and Retest
Next Executive Decision	Approve or reject affected integrations only after the defined acceptance criteria are demonstrated